

Operations Project Authorization Form

Approved on March 28, 2024 EPAC

[Link to Meeting Minutes](#)

Project Title: Hyde Park Battery Energy Storage System	Project Number: 21334
Date Prepared: 03/25/2024	Company: Eversource EMA
Organization: System Planning	Class(es) of Plant: D SS
Project Initiator: Mariam Chatha	Project Category: Stations - Batteries
Managing Organization: Project Mgt and Construction	Schedule ID#: 80089926
Project Manager: Adam Kelley	Project Type: Specific
Project Sponsor: Lavelle Freeman	Capital Investment part of original Op Plan: Yes
Estimated in service date: September 2027	Emergency Related Request? No
Eng./Constr. Resources Budgeted? Yes	Facility Type (check all that apply): ☐ PTF ☐ non-PTF ☐ local PTF ☒ Distribution ☐ N/A (General Plant)
EJ Community: Yes	Authorization Type: Partial Funding
Total Capital Request: \$500K	

[1] Executive Summary

This is a partial funding request of \$500K to fund the continuation of engineering and legal counsel for a 10 MW/20 MWh utility-scale battery storage system (BESS) at 427 Cummins Highway near Hyde Park Substation #496 in Hyde Park, MA.

This project had initial funding of \$200K approved in PowerPlan on June 16, 2021, and a revised 21334 - Hyde Park BESS – Battery Energy Storage System Solution, Rev 1 IFR for \$300K approved in PowerPlan on May 25, 2023. The initial funding was used to complete conceptual level engineering and estimates and development of a partial funding Project Authorization Form (PAF) and Solution Selection Form (SSF). The partial funding PAF and SSF were presented to the Solution Design Committee (SDC) on February 15, 2023 to fund the preliminary engineering, siting, and permitting for the BESS. The PAF and SSF were conditionally approved. This current partial funding PAF is requested to continue engineering and siting until approval of the partial funding PAF (tentatively in April 2024) for the completion of preliminary engineering. This PAF is requesting approval for an additional \$200K, and a new total of \$500K for the project. EPAC full funding approval is expected in August 2025.

Form Template Revised 03/13/2024

APS 1 – Project Authorization Policy Issued 12/31/21 Rev. 6

Policy Sponsor: EVP and CFO

JA-AM-2001-A Rev. 7

The primary objective of the proposed project is to help relieve the heavy loading condition of Hyde Park Substation #496.

Hyde Park Substation #496 is 90% above the substation load carrying capability (LCC) and will continue to increase each year. A mid-term solution to bring additional capacity to the area is needed before a long-term solution will be in service. The long-term solution project is 21192 - Hyde Park-Dorchester Area Supply Long-Term Plan (Reference 1); a partial funding PAF was approved at EPAC on August 14, 2023 but will be resubmitted to accommodate a 15-year station plan versus a 10-year plan. The expected in service date for the long-term project is December 2031. The proposed BESS is the proposed mid-term solution.

The project 22345 – 23375 – Hyde Park-Dorchester Area Plan: Land Acquisition (Reference 3) was approved by EPAC on September 11, 2023 for the amount of \$6.275M and the property was purchased in October 2023. The property at 161-191 New England Avenue, Boston, which has been purchased by project 22345-23375 to support project 21192 – Hyde Park-Dorchester Area Supply Long Term Plan.

[2] Project Costs Summary

Note: Dollar values are in thousands and rounded to nearest thousand

Project Number: 21334

Project Title:

D- Station

Template version;
rev25

Line item Category	Prior Authorized	Actuals to Date	2024 to Go	Total
1. ROW / Easements / Land purchases				
2. Environmental Approvals / Permits		\$6	\$3	\$8
3. Outreach		\$12	\$3	\$15
4. Siting Approvals / Permits		\$12	\$9	\$21
5. Engineering / Design		\$17	\$7	\$24
6. Materials (Eversource purchased)		\$2		\$2
7. Construction (incl matl's by contractors)		\$11		\$11
8. Testing / Commissioning				
9. PMT Project Mgmt Team		\$88	\$3	\$91
10. Removals				
11. Other		(\$1)		(\$1)
12. Risks				
Project Directs Total w/Risks	\$300	\$148	\$23	\$170
13. Indirects/Overhead		\$222	\$47	\$269
14. AFUDC		\$30	\$30	\$60
Project Total - Baseline Budget	\$300	\$400	\$100	\$500
15. Contingency				
#N/A				
Project Total - Funding Approval Level	\$300	\$400	\$100	\$500
16. Reimbursables/Customer Contribution				
#N/A				
Project Total (less reimbursables)	\$300	\$400	\$100	\$500
Total O&M				
TOTAL REQUEST	\$300	\$400	\$100	\$500

Detail other costs, project risk items and their allocations:

Risk	Allocation
Other costs (Mileage, Property taxes)	(\$1K)
1. *Risks will be accounted for in the Full Funding Request	\$
2.	\$
3.	\$
4.	\$
5.	\$
6.	\$
7.	\$
Total	\$

Explain unique payment provisions, if applicable:

N/A

Identify and detail any unique O&M impacts resulting from this project:

The project will incorporate new equipment not normally maintained by Eversource, including battery modules, inverters, and other energy storage system specific equipment. Contractual language and requirements will be placed in the Battery Energy Storage System Vendor Request for Proposal to require training of Eversource personnel, material warranty and spares, and O&M manuals for all non-standard equipment.

[3] Technical Justification

Project Need, Objective and Justification

Project Need

Hyde Park Substation #496 2021 peak load was 157MVA, which exceeded the 2021 Load Forecast by 7MVA. It is 108% or 11MVA above Substation firm capacity of 146MVA; 98.7% of the Substation Load Carrying Capability (LCC) of 159 MVA. The firm capacity is the capacity based on the loss of a Hyde Park Substation transformer, and the LCC is the firm capacity in addition to the amount of load that can be transferred to nearby substations utilizing distribution circuit ties. In 2021, 7.7 MVA was transferred permanently from Hyde Park Substation #496 after the Substation peaked for that year. The 2022 post load transfer peak load was still 152MVA, 104% of the Substation firm capacity of 146MVA and 95.6% of the Substation LCC of 159 MVA. About an additional 5 MVA of load was transferred in 2022 after the Substation peaked for that year. In 2023, the Substation was loaded to 97% of its firm capacity. A total of 13 MW of load was transferred to nearby substations using existing field switches. However, even with these load transfers, the Hyde Park Substation #496 is forecasted to be about 95% of the Hyde Park Substation LCC by 2024.

Load transfers are not a feasible option because they reduce the existing circuit transfer capability, increase the existing distribution lines' exposure, and reduce the spare capacity at the receiving substation. Building a new substation to address this thermal constraint is expected to be completed in December 2031 (or later; Project 21192) and is a long-term solution to address the capacity constraint in the area. Therefore, a near term mitigation plan is needed to relieve this thermal substation constraint until the long-term solution, 21192 - Hyde Park-Dorchester Area Supply Long-Term Plan referenced above is in service.

Project Objective

The objective of the project is to alleviate Hyde Park heavily loaded condition, to prevent the Substation from being overloaded before the long-term solution, potentially a new Substation installation, can be materialized in the area via project 21192 - Hyde Park-Dorchester Area Supply Long-Term Plan. The property at 161-191 New England Avenue, Boston has been purchased by project 22345-23375 to support project 21192 – Hyde Park-Dorchester Area Supply Long Term Plan.

Four (4) 2.5MW/5 MWh battery units will be installed at 427 Cummins Highway, each connected to a Hyde Park 13.8kV distribution feeder. The batteries would be dispatched with a peak load shave functionality. They will charge during the light load and discharge during the peak load in order to lower the substation peak. The battery system will be able to operate through multi-day peak load events.

By July of 2027, the capacity to be delivered by the project (21334) will need to be delivered by other means. Delays may include but are not limited to procurement, siting, final engineering,

construction. Those options may include deployment of spot generation, voltage reduction, and load shedding during peak load hours during the summer timeframe typically from 18:00 PM to 20:00 PM. Based on the current 2023 forecast, which may be subject to change in the future, if the BESS is not installed by July 2027, two (2) generators of 1 MW capacity each would be needed in 2027 and 2028 and two (2) generators of 2 MW capacity each would be needed in 2029 until the BESS is installed. 8 MW of generation would be needed in the worst-case scenario in which the BESS does not get installed until the station is at 100% of LCC. Spot generation would be required to supply load during peak load hours and can remain on standby for the remainder of the time. System Planning will monitor the annual forecast that may shift the plan laid out above.

Project Scope

The Hyde Park BESS project consists of a 10MW/20MWh energy storage system. The Project will be located at 427 Cummins Highway in Boston on Eversource-owned property. The Hyde Park BESS will be interconnected to the Eversource electrical grid. The BESS shall be comprised of a system of battery racks, inverters, control system, pad mounted transformers, and switchgear. The BESS will be located and interconnected to Eversource's existing Hyde Park 13.8kV distribution circuits infrastructure adjacent to the BESS facility.

- The BESS facility is proposed to interconnect to the following 13.8kV circuits supplied from Hyde Park substation #496:
 - 441-1401H
 - 362-1207H
 - 362-1209H
 - 496-H1XY
- Required Distribution System improvements include:
 - Extend distribution circuits 5,465 feet up to and into the Cummins Highway property for the BESS interconnection. Scope of work includes the installation of three (3) manholes, conduit, and associated cable.
- BESS Preliminary Scope of Distribution Work
 - Installation of (3) concrete manholes with inside dimension of 6'x10'x8'
 - Installation of (1,270') 6-5" PVC type EB conduit encased in concrete from MH12362 to BESS via three (3) new manholes.
 - Primary Underground Cable: Removal (13,065') of 15kV primary cable: 350/3, 3-350 FS, 3-500 FS, etc. (37 sections)

- Primary Underground Cable: Installation (16,395') of 3-700 FS 15kV primary cable (47 sections)
- Interconnection to Circuit 496-H1XY
 - Cut and remove X-Y main from MH18478 to MH12361 and replace with 2,950 feet of 3-700 FS cable (9 sections) and make joints.
 - Install 1,270 feet of 3-700 FS cable (4 sections) from MH12362 to new BESS and make joints.
 - Note: One (1) single three-way joint in MH18478 and three (3) single multiple joints in MH12361.
- Interconnection to Circuit 441-1401H
 - Install new cable from MH12365 to MH12362 with 1,200 feet of 3-700 FS cable (3 sections) and make joints.
 - Install 1,270 feet of 3-700 FS cable (4 sections) from MH12362 to new BESS and make joints.
- Interconnection to Circuit 362-1207H
 - Cut and remove cable from MH15729 to MH12361 (5 sections).
 - Install 1,810' feet of 3-500 FS cable (5 sections) from MH15729 to MH12361 and make joints.
 - Install 1,270 feet of 3-700 FS cable (4 sections) from MH12362 to new BESS and make joints.
- Interconnection to Circuit 362-1209H
 - Cut and remove cable from MH2434 to MH12358 (14 sections).
 - Install 5,355 feet of 3-700 FS cable (14 sections) from MH2434 to MH12358 and make joints.
 - Install 1,270 feet of 3-700 FS cable (4 sections) from MH12362 to new BESS and make joints.

The \$500K partial funding being requested will be used to progress the project toward preliminary design until the conceptual level partial funding approval. Partial funding will also be for the following:

Distribution System Planning

- Coordinate with the various stakeholders to carry the project up until conceptual level partial funding approval which advances the project to preliminary design.

Capital Project Engineering

- Coordinate the overall engineering effort for the below external and internal disciplines.
 - Battery Energy Storage System Vendor
 - Substation Engineering
 - Protection & Control Engineering
 - Telecom Engineering
 - Distribution Engineering
- Finalize technical specifications for the Battery Energy Storage System Vendor Request for Proposal. Engineering has decided to bring on an engineering vendor to support this technical spec development.
- Procurement of Battery Energy Storage System vendor to develop conceptual and preliminary packages required for the installation and interconnection of the battery system as outlined above.

Project Management

- Overall project planning and scheduling
- Project reporting, document control and management, and financial projections
- Battery Energy Storage System Vendor procurement coordination

Siting

- Coordinate and lead siting effort which includes pre-filing meetings with the City of Boston for the Article 80 Small Project siting path. Coordination with Eversource legal on the onboarding of outside legal counsel to assist with the submittal for Small Project Review, along with assistance throughout the duration of the permitting process and securing approval at the Boston Zoning Board of Appeals. Siting Filing will occur after Full Funding Approval.

Project Services

- Initiate the community outreach and engagement strategy required for the project through the Project Services and Community Relations teams. This outreach strategy has been reviewed and communicated with the City of Boston and Boston Planning and Development Authority and has been aligned and coordinated with the outreach strategy for project 21192 – Hyde Park-Dorchester Area Supply Initiative.

Equity and Environmental Justice (EJ)

- Identify EJ Communities and populations relevant to the Project. Identify languages spoken in stakeholder communities to enhance strategic communication with multilingual customers. Identify and communicate with communities and stakeholders in a broad geographic area to determine historical engagement and determine potential Project or policy impact.

Environmental

- Project support and preparation of regulatory agency required permits identified on the Environmental Affairs Department project review form.
- Coordinate required meetings with Department of Environmental Protection for utilization of the Cummins Highway parcel for development as it is currently a capped landfill (location of proposed BESS)

Background

In 2021, Hyde Park Substation #496 exceeded the 2021 Load Forecast by 7MVA. It is 108% of Substation firm capacity of 146MVA, and 98.7% of Substation LCC of 159MVA. In 2022, 7.7MVA was transferred permanently off Hyde Park Substation #496. The 2022 post load transfer peak load was still 152MVA, 104% of the Substation firm capacity of 146MVA and 95.6% of the Substation LCC. In 2023, the substation was loaded to 97% of its firm capacity after additional load transfer.

Below is the Hyde Park Substation #496 ten (10) year load forecast and percentage of LCC.

	Station		2024	2025	2026	2027	2028	2029	2030	2031	2032
		Forecast Load	150.83	150.86	150.97	151.09	151.36	151.81	152.43	153.41	155
496	Hyde Park	Load/LCC	94.9%	95.5%	95.6%	96.2%	96.4%	97.3%	97.7%	98.3%	100%

A 13MW load transfer was implemented by using existing field switches. However, such transfer reduces the existing circuit transfer capability, increases the existing distribution lines' exposure, and reduces the spare capacity at the receiving substation. Therefore, this cannot be used as a long-term solution.

Additional capacity is required for Hyde Park to relieve the heavily loaded condition. Building a new substation to address this thermal constraint will take a number of years and is a long-term solution to address the capacity constraint in the area. Therefore, a near-term mitigation plan was developed to relieve this thermal substation constraint until the long-term plan is in service. A BESS or construction of distribution lines from a nearby Substation to the area was considered.

[4] Alternatives Considered

This project was presented to and conditionally approved at the February 15, 2024, SDC meeting. It is anticipated the project will have final approval in April 2024.

Alternative 1 (preferred alternative):

Install four (4) 2.5MW/5MWH Battery Energy Storage System with a total capacity of 10MW/20MWH at Cummins Highway.

With the proposed BESS installation and no distribution load transfers, the Hyde Park Substation #496 load is expected to relieve around 92% of the Substation LCC by 2028 as follows:

	Station		2024	2025	2026	2027	2028	2029	2030	2031	2032
496	Hyde Park	Forecast Load	150.83	150.86	143.97	144.09	144.36	144.81	145.43	146.41	148
		Load/LCC	94.9%	95.5%	91.12%	91.78%	91.95%	92.83%	93.22%	93.85%	95.48%

The BESS is proposed to interconnect to four (4) 13.8kV distribution feeders, 496-H1XY, 441-1401, 362-1209H and 362-1207H as shown in Figure 1 below. The proposed BESS system is expected to perform peak load shave functionality to relieve the substation heavy loading condition.

Four 2.5MW/5MWH batteries are chosen to allow for bus and feeder diversity as well as battery diversity. As shown in the figure below, each battery is tied to a feeder supplied from a different Hyde Park 13.8 kV Bus. If there is a fault on either the bus or feeder, the BESS can connect to the remaining buses and feeders. Similarly, if there is a failure in one of the batteries, there would still be 7.5MW/15MWH capacity available from the batteries.

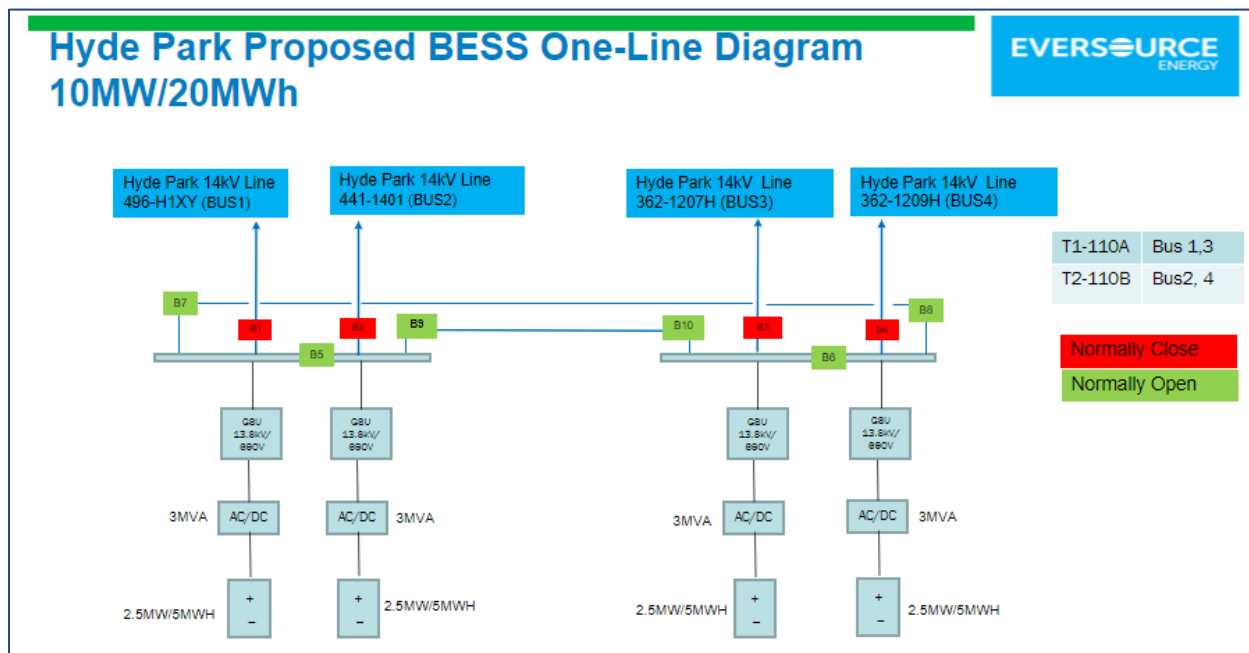


Figure 1. BESS One line Diagram

Advantages:

- BESS can help relieve heavily loaded Hyde Park Substation #496.
- Construction is confined to a single site where the batteries can each interconnect to four (4) different 13.8kV Hyde Park distribution feeders and provide substation bus diversity.
- Can be completed in a minimum of three (3) years.
- This option has the least impact on the community.
- Shorter timeline for Outreach as compared to other alternatives since discussions with the City of Boston are already underway.
- Expected timeline for outreach, which has already commenced is approximately 12 – 18 months before a filing, which would likely be around Q2/Q3 of 2024 and continue ongoing thereafter.
- There has been early involvement with the City of Boston for this alternative, which helps to smooth the path for siting the BESS with City support and reduces opposition.
- After the long-term project (21192) is in service, the BESS can be redeployed to other locations where it can be useful.

Disadvantages:

- This alternative will require a City of Boston Article 80E Small Project Review and a variance with the Boston Zoning Board of Appeals. It is not known if previous utility only projects have been approved through this process but a review of previous projects by the Siting team has indicated an average timeframe for approval of 18 months.

- If the Article 80E and Zoning Relief process is not successful, a 40A zoning exemption would be required from the Department of Public Utilities (DPU) which would add significant time to the project.
- There has been local and neighborhood opposition for a proposed housing development on an adjacent parcel and similar opposition and concerns are expected for the BESS development. Project Services and Community Relations have been engaging with the City on this issue and will be performing targeted community outreach for this project and the Hyde Park solutions as a whole (Project 21192).

Hyde Park and surrounding downtown Boston parcels include areas where the majority of the population is a minority and speak English as a second language (EJ Communities). Adding more capacity to the system helps provide reliable service to the population.

Conceptual (-25%/+50%) BESS Cost Estimate	\$44.379 million
Conceptual (-25%/+50%) Distribution Scope Cost Estimate:	\$ 4.793 million
Conceptual (-25%/+50%) Total BESS Cost Estimate	\$49.172 million total

Hyde Park BESS Load Flow Analysis

The firm capacity of the Hyde Park Substation #496 is 146 MVA while the LCC is 159 MVA. The LCC decreases incrementally by a slight amount every couple of years.

Assuming the peak day load profile in the Figure 2 below, the four (4) 2.5MW/5 MWh battery units can reduce the peak load of the Hyde Park Substation #496 to approximately 95% of the LCC in year 2032 or approximately 148 MVA. The BESS would be interconnected to four (4) 13.8kV distribution feeders, 496-H1XY, 441-1401, 362-1209H and 362-1207H. One (1) 2.5MW/5 MWh battery will be connected to each of the four (4) feeders. The project is designed to meet the ten (10) year forecast, where the highest load is in 2032.

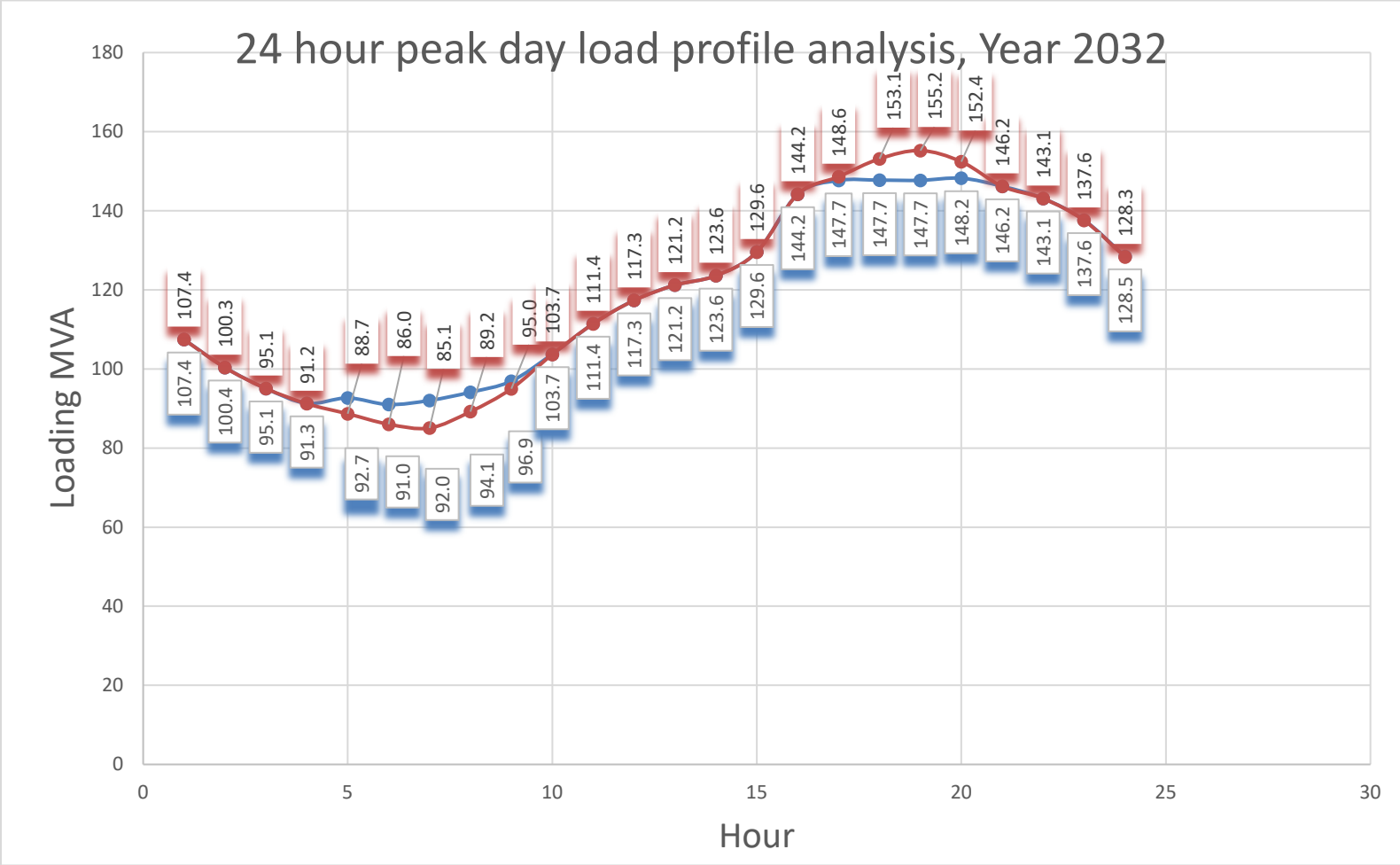


Figure 2 Hyde Park Substation Loading

In the above graph (Figure 2), the red line represents the Hyde Park Substation #496 load without any batteries installed and the blue line represents the substation load with the four (4) batteries installed. The graph below (Figure 3) shows the 2032 24-hour peak day loading on each of the four (4) feeders that the BESS is interconnected to and assumes the BESS is operational.

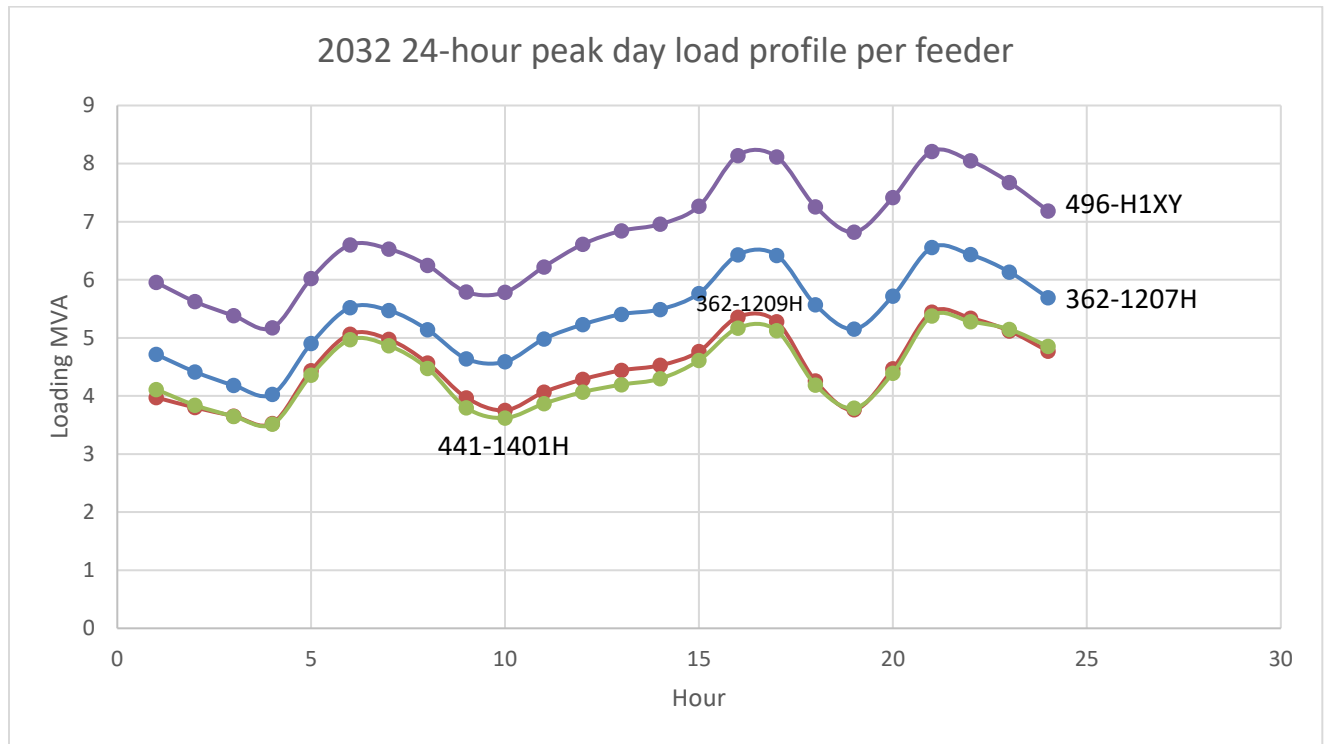


Figure 3 2032 Hyde Park 13.8kV Loading

As shown by Table 1 below, the circuit loading is below 80 percent of the circuit ratings when the battery is charging from 5 AM to 9 AM and acting as a load.

	Feeder_362_1207H	Feeder_362_1209H	Feeder_441_1401H	Feeder_496_H1XY
5:00 AM (MVA)	4.9	4.4	4.4	6.0
6:00 AM (MVA)	5.5	5.1	5.0	6.6
7:00 AM (MVA)	5.5	5.0	4.9	6.5
8:00 AM (MVA)	5.1	4.6	4.5	6.3
9:00 AM (MVA)	4.6	4.0	3.8	5.8
80% of Circuit Rating (MVA)	6.72	6.72	8.72	7.36
Circuit Rating (MVA)	8.4	8.4	10.9	9.2

Table 1 2032 Hyde Park 13.8kV Feeder Loading With BESS

During the annual planning forecast, System Planning will be monitoring and modifying the battery schedule based on the Electric Vehicle charging profile and its load on the Hyde Park distribution circuits.

The below graph (Figure 4) shows the resultant load profile for the substation during the peak load day under contingency assuming 13.17 MVA of distribution load transfers and the battery is

operational. This decrease of load to 141 MVA is due to a load transfer of 13.17 MVA which occurs during N-1 condition through distribution feeder ties and the decrease to 133.7 at 19:00 is due to the BESS.

The red line represents the Hyde Park Substation #496 load without any batteries installed and the blue line represents the Substation load with the four (4) batteries installed.

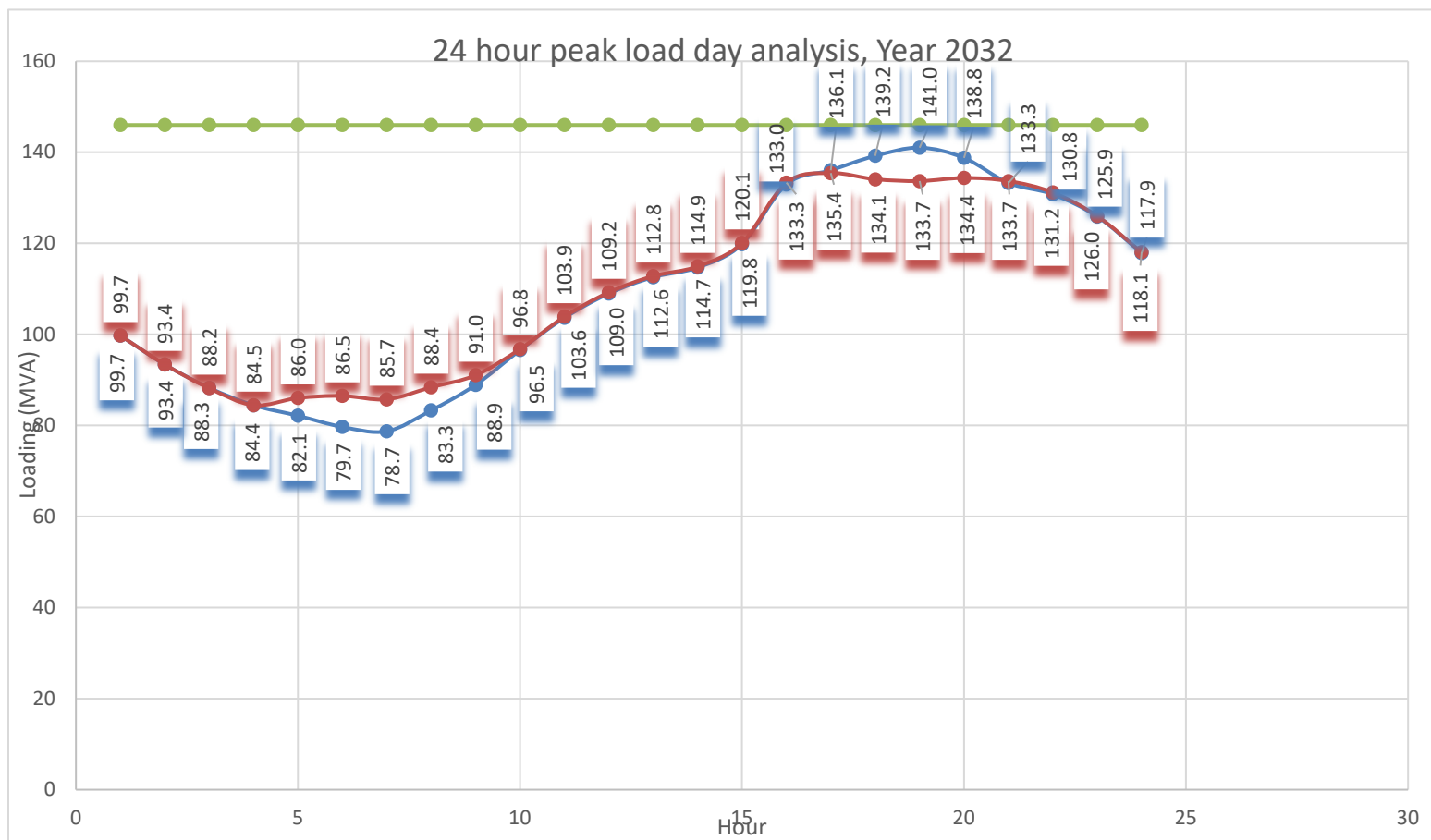


Figure 4 2032 Hyde Park Substation 24 Hour Loading Under N-1

Below is a graph (Figure 5) showing circuit loading for each of the four (4) feeders under a Hyde Park Substation transformer contingency with the BESS connected to each of the four (4) feeders. It shows the 24-hour peak load day profile for the four (4) selected feeders in MVA based on the 2032 modeled loads. The batteries act as a load and charge from 04:00 AM to 09:00 AM and discharge from 05:00 PM to 09:00 PM to decrease the Substation peak load.

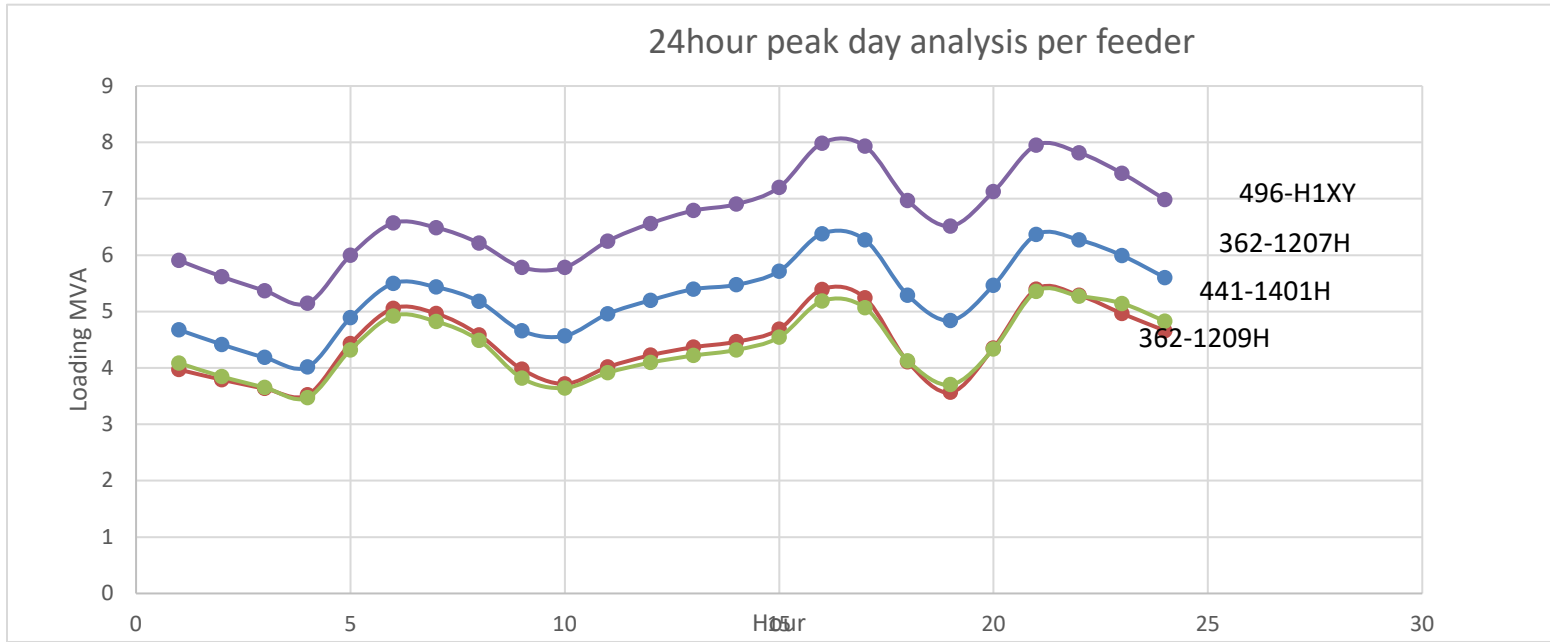


Figure 5 2032 Hyde Park 13.8kV 24 Hour Feeder Loading Under N-1

As shown by Table 2 below, the circuit loading is below 80 percent of the circuit ratings when the battery is charging from 04:00 to 09:00 and acting as a load.

	Feeder_362_1207H	Feeder_362_1209H	Feeder_441_1401H	Feeder_496_H1XY
5:00 AM	4.89	4.43	4.32	5.996
6:00 AM	5.50	5.06	4.92	6.57
7:00 AM	5.43	4.96	4.82	6.49
8:00 AM	5.18	4.59	4.49	6.21
9:00 AM	4.66	3.97	3.82	5.783
80% of Circuit Rating (MVA)	6.72	6.72	8.72	7.36
Circuit Rating (MVA)	8.4	8.4	10.9	9.2

Table 2 2032 Hyde Park 13.8kV Feeder Loading with BESS Under N-1

Alternative 2:

Bring additional capacity to Hyde Park area by constructing two (2) 13.8kV distribution lines from Needham Substation #148

The Needham Substation #148 (Needham MA) firm capacity is 142 MVA while the projected peak load in the next ten (10) years is 88 MVA.

This alternative is based on an assumption that the new dedicated lines from Needham Substation #148 would run 100% in an underground path, not currently occupied by common duct. The two (2) new lines would run from Needham Station #148 to nearby Hyde Park 13.8kV circuits, 496-H3 and 496-H6. The need to provide these two (2) circuits would require:

- 19,600 feet of 9-6" PVC conduit
- 52 manholes
- 360 feet of 6-6" PVC conduit
- 350 feet of 2-5" PVC conduit
- 21 (55-foot poles)

The following cable work would be needed for the circuit that would relieve Hyde Park 13.8 kV circuit 496-H3:

- 19,900 feet of 3-700FS
- 2,450 feet of 477 ACSR
- 2 new Reclosers
- 2 new Underslung
- 2 new 45-foot poles

This new circuit would offload 7MVA from this 496-H3 circuit.

The following cable work would be needed for the circuit that would relieve Hyde Park 13.8kV circuit 496-H6:

- 25,900 feet of 3-700FS
- 2,300 feet of 477 ACSR
- 2 new Reclosers
- 2 new Underslung

This new circuit would offload 6.21MVA from this 496-H6 circuit.

There is some load that would still be kept on 496-H6 and 496-H3 just for cable monitoring purposes so that the cable will have readings and can therefore be utilized under a fault via a tie switch. If there is no load left on the feeders, they cannot be monitored to determine whether they can be utilized as distribution ties.

With this alternative, the total possible capacity on each circuit is 11.5MVA per circuit.

Figure 6 below shows the high-level route (dashed black line) from Needham Substation #148 to Hyde Park 13.8kV circuits 496-H3 and 496-H6.

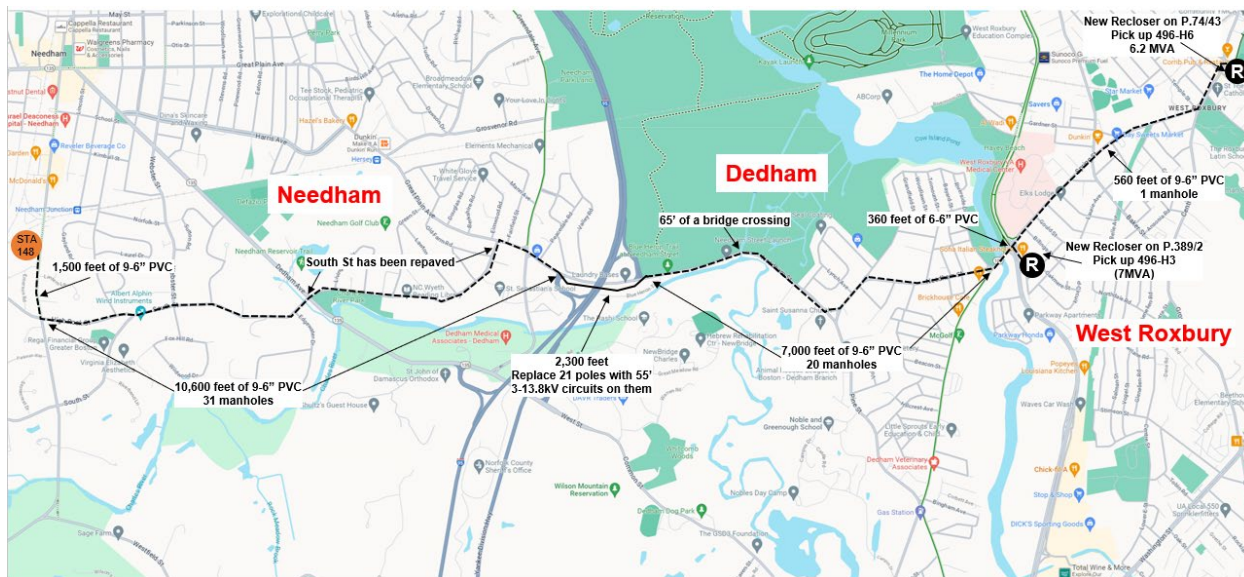


Figure 6. Needham 13.8kV Feeders

Alternatively, the option of running the 13.8kV circuits through an overhead right-of-way (ROW) to connect to the two (2) Hyde Park 13.8kV circuits was also analyzed. This option was not feasible because the railroad easement along the path would limit the number of circuits that can run along the route; the width between the tracks and edge of easement is also too limited to accommodate two (2) 13.8kV circuits in some locations.

Advantages:

- This alternative would relieve the heavily loaded Hyde Park Substation #496.
- A filing with the Energy Facilities Siting Board (EFSB) or DPU is not anticipated since this is a distribution related solution.

Disadvantages:

- Construction will be crossing multiple towns, busy highways, and bridges, having an adverse impact on the community.
- This alternative will either take the same amount of time as Alternative 1 or more.
- Due to the extensive in-street work required for this Alternative, Project Services and Community Relations would have to undertake substantial community outreach and dialogue with City officials, which has not commenced. No outreach has been done with municipalities on this alternative. This would require multiple meetings with City staff, elected officials, community-based organizations, etc. Expected timeline for outreach is approximately 12 – 24 months before a filing and continue ongoing thereafter. Extensive permitting would also be needed to cross multiple towns.
- Opposition in Needham is expected since a project was just completed through Needham. If residents object or create organized opposition, significant delays to approvals are expected.

- Routing options would need to be discussed with cities and other stakeholders.
- Rights would need to be obtained from the towns/cities to install the conduits. Grant of location would be needed for Needham, Dedham and Boston.
- Permits would be needed to cross Route 128 and the bridges, and other entities such as Massachusetts Water Resources Authority or Department of Conservation and Recreation This could take up to a year.
- The following caveats pose as disadvantages as well and can affect the cost estimates for this alternative:
 - South St in Needham has been recently paved, so it is uncertain whether curb to curb paving would be required.
 - Two (2) overhead lines will be installed to cross Route 128. This may result in increased labor costs and decreased reliability of these circuits.
 - Conduit/pipes need to be installed over a bridge crossing at the Dedham/Needham line.
- Expect significant mitigation requests for roadwork restoration, such as curb-to-curb/gutter-to-gutter paving, Americans with Disability Act ramp restoration, new sidewalks, and/or traffic calming measures like speed humps, and other community benefits, which could include monetary payments for trees or other types of added green space.
- Depending on the extent of work required at the Needham and Hyde Park substations (not yet known), zoning relief may be required from the Town of Needham and the City of Boston.
- Based on the current forecast, this alternative would not remain as useful after the long-term project, a new 115kV/13.8kV substation is in service to relieve load from Hyde Park and surrounding substations.

Conceptual (-25%/+50%) Cost Estimate:

\$46.473 million total

Alternatives Summary

Although the cost for the Alternative 2 is lower than the preferred alternative, it is critical to note the hurdles and timing delay that would be faced if this alternative were to be implemented as explained above. Opposition in Needham is expected since a project was just completed through Needham. If residents object or create organized opposition, significant delays to approvals are expected. There are also potential hurdles involved in gaining rights and permits for Alternative 2. Alternative 2 may take significantly longer than the BESS to implement as explained above. Outreach for the BESS Alternative 1 has already been ongoing which helps to smooth the path for siting the BESS with city support and reduces opposition whilst there has been no outreach for Alternative 2.

Two of the feeders that the BESS would connect to would also benefit from the BESS, Circuit 496-H1XY supplied from Bus 1 historically runs 80-90% of its summer normal ampacity in the

summer. The circuit 362-1209 supplied from Bus 4 historically runs 60-90% of its summer normal in the summer. The BESS would help to reduce the loading on these circuits.

The BESS can be redeployed to other stations where it can be useful after the long-term project is in service. However, based on the current forecast, Alternative 2 would not be as useful after the long-term project or new 115/13.8 kV substation is in service to relieve load from Hyde Park and surrounding substations.

[5] Project Schedule

Milestone	Scheduled Completion Date
IFR Approval	June 2021 May 2023 (Rev 1)
SDC Approval (for applicable projects)	February 2024
Conversion of Tx development costs, 183 to 107 (applicable to projects that require SSF approval and/or ISO preferred alternative selection)	N/A
EPAC Partial Funding Approval (for applicable projects)	March 2024
Preliminary Design Completion (or latest pre-full funding design phase)	June 2025
EPAC Full Funding Approval	August 2025
Siting Approval (for applicable projects)	October 2026
Start of Construction	November 2026
Project In-Service Date (ISD)	September 2027
Project Need Date (for applicable projects)	Present
ISO Related Milestones: (for applicable projects)	
• ISO Preferred Alternative Selected	N/A
• PAC Presentation	N/A
• TCA Submittal	N/A

P6 schedule is attached as attachment B.

[6] Regulatory Approvals

The list of regulatory approvals will be finalized after Full Funding per direction from SDC; expected permits/approvals include:

- The BESS (Alternative 1) is expected to require Article 80E Small Project Review and a variance from the Boston Zoning Board of Appeal. Siting, with the assistance of outside legal counsel, would file for and obtain the required zoning relief.
- If the City of Boston is unsupportive of providing zoning relief, then Siting would file for a zoning exemption under M.G.L. Chapter 40A §3 from the Department of Public Utilities (DPU) (Reference 4). The expected timeframe for a zoning exemption would be about 18 months to 24 months from date of filing to a decision.
- Necessary environmental permit approvals include:
 - Environmental Protection Agency (EPA) National Pollution Discharge Elimination System (NPDES) Construction General Permit (CGP) and Stormwater Pollution Prevention Plan (SWPPP)
 - Requires development/engineering of structural stormwater management system for facility and development of Operations and Maintenance (O&M) Plan for long term operation of stormwater management system, including semi-annual inspection, maintenance, and record keeping requirements
 - Requires regular construction phase erosion and sediment control inspections and reporting
 - Massachusetts Department of Environmental Protection (MassDEP) Landfill Post-Closure Use Permit (PCUP)
 - Requires both construction phase and long-term operation inspection, monitoring, and reporting requirements
 - May require installation of sophisticated gas monitoring system inside facility buildings to send real-time alert to Operations group for any buildup of combustible and/or toxic gasses within enclosed spaces
 - Soil and Groundwater Management Plan
 - Dictates regulatory requirements for construction phase handling, management, and disposal of contaminated soils and groundwater regulated under the Massachusetts Contingency Plan
 - Spill Prevention, Control, and Countermeasure Plan (SPCC) and Tier II Reporting to EPA, MassDEP, Massachusetts Emergency Management Agency Local Fire Dept / Emergency Response Agency
 - Required for any new oil filled equipment with capacity >1,320 gallons.
 - Required for onsite storage/use of large quantities of hazardous materials (batteries)

EJ Community Considerations:

	EJ Mapping	Identify EJ Communities and populations relevant to the Project, through creation of an EJ Map. EJ Map will identify, by Census Block: • Neighborhoods with English isolation • Low-income areas • Areas with concentrations of racial/ethnic minorities • Areas with vulnerable populations (including those with high existing environmental burdens) • Areas with combinations of any of the above • Distressed municipalities and any sub-areas with concentrations of EJ communities • Areas containing sensitive infrastructure or critical facilities (e.g., public housing, nursing/elder care homes or facilities, parks, hospitals, schools) • Areas containing high residency of low to moderate income renters
	Stakeholders Mapping & Engagement	Identify and communicate with communities and stakeholders in a broad geographic area to determine historical engagement and determine potential Project or policy impact, including local community-based organizations and leaders, state-wide NGOs and Indigenous tribes.
	LANGUAGE TRANSLATION SERVICES	Identify languages spoken in stakeholder communities to enhance strategic communication with multilingual customers. Language identification data shall be used to inform and secure interpreters, generate appropriate translations of materials, conversations, dialogue, and in-community events.

	PUBLIC ENGAGEMENT	Host public engagement processes, including through non-traditional and community-based communication channels, to solicit public input. This may include, as applicable: • Determining the level of community interest • Conducting analysis of stakeholders who will be immediately impacted by the delivery or implementation of the project/services • Creating and implementing engagement & communication strategies • Hosting in person and or virtual public feedback sessions to clearly define project/services scopes of and timelines • Identify additional stakeholders to attend future public engagement activities and follow-up on feedback received
	Feedback	Establish a database to track all outreach events, interactions, materials, communications received from communities, individuals, and organizations; record all responses, all changes/modifications to projects or programs, and all remediation, mitigation, and community benefit arrangements made to or with respect to relevant EJ communities.

[7] Risk and Risk Mitigation Plans

Contingency:

Contingency will be included in the full funding PAF request.

Risk Mitigation:

Risks will be identified using the partial funding and included in the full funding PAF request.

[8] References

References
1. 21192 -Hyde Park -Dorchester Area Supply Initiatives Partial Funding PAF, approved 03/02/2023
2. 22345 – 23375 – Hyde Park-Dorchester Area Plan: Land Acquisition, approved 09/11/2023
3. 21334 - Hyde Park BESS – Battery Energy Storage System Solution, Rev 1 IFR, approved 05/09/2023
4. M.G.L. Chapter 40A §3 from the Department of Public Utilities (DPU).
5. 21334 - Hyde Park BESS - SSF

[9] Attachments (One-Line Diagrams, Images, etc.)

Attachment	Included in Submittal?
A. Cost Estimate	YES
B. P6 Schedule	YES
C. Constructability Review	NO

EVERSOURCE

Project Authorization Form

INSTRUCTIONS:

It is the responsibility of the initiator to contact the subject matter experts in the listed area disciplines to determine if the project considerations contained in this list are applicable to their project. They should fill out the checklist and determine a transition plan for the purpose of project execution. See the PAF Guide for additional instruction.

Project Checklist - Transmission & Substation Capital Project	
Project Name: Hyde Park BESS – Battery Energy Storage Solution	Project Number: 21334
Facility Type: ☐ BPS ☐ BES ☐ PTF ☐ local PTF ☐ non-PTF ☐ CIP ☒ Distribution ☐ General Plant ☐ CONVEX ☐ SCADA	
PLANNING	
Is a NX-9 required?	No
Is an ISO-NE PAC presentation required?	No
ISO-NE Presentation Date (if completed):	N/A
Cost in ISO-NE Presentation (\$M) (if completed):	N/A
Is a PPA required?	Yes
ISO-NE Approval Date (if completed):	TBD
Is a TCA Application Required?	No
TCA Submittal Date (if submitted):	N/A
Cost in TCA Submittal (\$M) (if submitted):	N/A
Enter ISO-NE RSP / Asset Condition ID, if assigned:	ES-23-LSP-032
PLANNING/PROTECTION & CONTROLS	
Are RAS/SPS/UVLs affected?	No
OPERATIONS	
Outage Required?	☐ Primary Equipment (Power Transfer) ☒ Secondary Equipment (P&C only) ☐ Outage Not Required
Do SCLL Conditions Exist?	No
Has an outage schedule been approved?	No
Are Operations & Maintenance procedures/training required?	Yes
STANDARDS	
Does the project include standard equipment and designs?	Yes
SUBSTATION ENGINEERING	
Does this impact Revenue Metering	No
Is preliminary short circuit/ breaker duty analysis required?	No
Are there any changes to the baseline audible noise?	Yes
Is there an impact to the existing ground grid?	No
Is a Transient Over Voltage (TOV) analysis required?	Yes

EVERSOURCE

Project Authorization Form

Project Checklist - Transmission & Substation Capital Project	
Project Name: Hyde Park BESS – Battery Energy Storage Solution	Project Number: 21334
P&C ENGINEERING	
OP-22 - Are PMUs and DDR required?	No
If BPS, is an NPCC Directory #4 presentation required?	No
Are ISO-NE OP-24 Appendix B updates necessary?	No
TRANSMISSION LINE ENGINEERING	
Are there any changes that affect the baseline EMF?	No
Are there any changes that affect the baseline EMI?	No
SITING	
Has Siting reviewed this project?	Yes
Name of Siting contact:	Eamon McGilligan
Is a Siting filing required? <i>(If yes, list in regulatory approvals section)</i>	Yes
PERMITTING	
Is there any permitting required? <i>(If yes, list in regulatory approvals section))</i>	Yes
PROJECT SERVICES (OUTREACH) and ROW WORK COORDINATION	
Has Project Services reviewed this project?	Yes
Name of Project Services contact:	Meredith Boericke
What is the level of outreach expected?	High
Has ROW Coordination reviewed this project?	Yes
Is this project coordinating with other existing scheduled work?	No
ENVIRONMENTAL JUSTICE COMMUNITIES	
Has the Equity and Environmental Justice team reviewed this project?	Yes
Name of Equity and Environmental Justice contact:	Alicine Francois
Is any part of this project in an Environmental Justice Community?	Yes
Was the Equity Framework Checklist completed?	No
Has enhanced outreach required per State Regulatory requirements been completed?*	No
INITIATOR	
Has a field constructability review been completed?	Yes

**Note: Enhanced Outreach will be developed by Full Funding.*

Project Checklist - Transmission & Substation Capital Project	
Project Name: Hyde Park BESS – Battery Energy Storage Solution	Project Number: 21334
INVESTMENT RECOVERY	
Does the project require development of an Investment Recovery plan?	No _____
COST ESTIMATING	
What team/firm prepared the cost estimate?	Estimate was prepared by the Eversource Estimating Team _____
Name of the person who prepared the estimate:	Peter Grenier _____
Was the estimate reviewed by Eversource Estimating?	Yes - Full Review _____

Project Number:

Project Title:

D- Station

PAF (dollars in thousands)

And on years e version; rev25

Line item Category	Prior Authorized	Actuals to Date	2024 to Go	2025	2026	2027	2028	2029	Total
1. ROW / Easements / Land purchases									
2. Environmental Approvals / Permits		\$6	\$3						\$8
3. Outreach		\$12	\$3						\$15
4. Siting Approvals / Permits		\$12	\$9						\$21
5. Engineering / Design		\$17	\$7						\$24
6. Materials (Eversource purchased)		\$2							\$2
7. Construction (incl matl's by contractors)		\$11							\$11
8. Testing / Commissioning									
9. PMT Project Mgmt Team		\$88	\$3						\$91
10. Removals									
11. Other		-\$1							-\$1
12. Risks									
Project Directs Total w/Risks		\$148	\$23						\$170
13. Indirects/Overhead		\$222	\$47						\$269
14. AFUDC		\$30	\$30						\$60
Project Total - Baseline Budget		\$400	\$100						\$500
15. Contingency									
#N/A									
Project Total - Funding Approval Level		\$400	\$100						\$500
16. Reimbursables/Customer Contribution									
#N/A									
Project Total (less reimbursables)		\$400	\$100						\$500
Total O&M									
TOTAL REQUEST	\$300	\$400	\$100						\$500

Breakout costs incl above

ST Labor		\$108	\$17						\$125
OT Labor									
Outside services - Labor		\$38	\$6						\$44
Materials (All materials incl. outside service purchased which is shown in constr. above)		\$2							\$2
Removals									
Other		-\$1							-\$1
Indirects		\$222	\$47						\$269
AFUDC		\$30	\$30						\$60
TOTAL CAPITAL REQUEST - with out Reimbursables	\$300	\$400	\$100						\$500

- Instructions:**
1. Input the project type info into the green cells on the Overall summary tab / answer escalation question
 2. Input direct costs (DOLLARS AND CENTS) and customer contribution (if applicable) in the green highlighted cells for the appropriate company which are separate tabs (green tabs)
 3. Input customer contribution as a negative number
 4. For D and Live work and station work have different tender rates so make sure you fill out the correct template section of each tab
 5. Input zero to green cells on overall summary tab. They then flow into the other tabs.

Template version: rev22		Template version: rev22		D - Station Work									
Direct Costs													
WBS Code	WBS Description	Prior Authorized	Actuals to Date	2024	2025	2026	2027	2028	Total				
Costs													
Eversource Internal													
1	ROW / Eversource / Land purchases												
2	Environmental Approvals / Permits												
3	Outreach		\$ 5,522	\$ 3,000									
4	Sting Approvals / Permits		\$ 8,686	\$ 3,000									
5	Engineering / Design		\$ 12,649	\$ 8,000									
6	Materials		\$ 3,509	\$ 2,000									
7	Construction		\$ 10,963										
8	Testing / Commissioning		\$ 63,811	\$ 1,000									
9	PM/ Project Mgmt Team												
10	Removals												
11	Other	Journal Entry Adjustment	\$ (1,037)										
12	Risks												
13	Contingency												
Outside vendor or contractor													
1	ROW / Eversource / Land purchases												
2	Environmental Approvals / Permits												
3	Outreach		\$ 2,633	\$ -									
4	Sting Approvals / Permits		\$ 11,155	\$ 4,000									
5	Engineering / Design												
6	Materials												
7	Construction												
8	Testing / Commissioning												
9	PM/ Project Mgmt Team		\$ 24,633	\$ 1,000									
10	Removals												
11	Other												
12	Risks												
13	Contingency												
		\$ -	\$ -	\$ 147,344	\$ 22,000	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 176,644
Escalation (if not included above)													
Escalation rates - Eversource Labor													
Escalation rates - Environmental Labor													
Escalation rates - Materials													
Eversource Internal													
1	ROW / Eversource / Land purchases	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
2	Environmental Approvals / Permits	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
3	Outreach	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
4	Sting Approvals / Permits	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
5	Engineering / Design	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
6	Materials	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
7	Construction	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
8	Testing / Commissioning	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
9	PM/ Project Mgmt Team	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
10	Removals	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
11	Other	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
12	Risks	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
13	Contingency	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Outside vendor or contractor													
1	ROW / Eversource / Land purchases	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
2	Environmental Approvals / Permits	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
3	Outreach	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
4	Sting Approvals / Permits	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
5	Engineering / Design	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
6	Materials	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
7	Construction	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
8	Testing / Commissioning	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
9	PM/ Project Mgmt Team	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
10	Removals	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
11	Other	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
12	Risks	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
13	Contingency	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Escalated costs													
Eversource Internal													
1	ROW / Eversource / Land purchases	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
2	Environmental Approvals / Permits	\$ -	\$ -	\$ 5,522	\$ 3,000	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
3	Outreach	\$ -	\$ -	\$ 8,686	\$ 3,000	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
4	Sting Approvals / Permits	\$ -	\$ -	\$ 12,649	\$ 8,000	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
5	Engineering / Design	\$ -	\$ -	\$ 5,509	\$ 2,000	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
6	Materials	\$ -	\$ -	\$ 3,509	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
7	Construction	\$ -	\$ -	\$ 10,963	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
8	Testing / Commissioning	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
9	PM/ Project Mgmt Team	\$ -	\$ -	\$ 63,811	\$ 1,000	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
10	Removals	\$ -	\$ -	\$ (1,037)	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
11	Other	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
12	Risks	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
13	Contingency	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Outside vendor or contractor													
1	ROW / Eversource / Land purchases	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
2	Environmental Approvals / Permits	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
3	Outreach	\$ -	\$ -	\$ 2,633	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
4	Sting Approvals / Permits	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
5	Engineering / Design	\$ -	\$ -	\$ 11,155	\$ 4,000	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
6	Materials	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
7	Construction	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
8	Testing / Commissioning	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
9	PM/ Project Mgmt Team	\$ -	\$ -	\$ 24,633	\$ 1,000	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
10	Removals	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
11	Other	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
12	Risks	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
13	Contingency	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
		\$ -	\$ -	\$ 147,344	\$ 22,000	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 176,644
PM/ Vendors													
1	Labor (PM/ROW/Env/Outreach/Sting)	\$ -	\$ 91,200	\$ 14,500	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
2	Labor (Eng/Const/Test)	\$ -	\$ 15,000	\$ 2,000	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
3	Materials	\$ -	\$ 2,500	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
4	Outside Services	\$ -	\$ 26,421	\$ 6,000	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
5	Other	\$ -	\$ (1,037)	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
6	Risks	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
7	Contingency	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
		\$ -	\$ 147,844	\$ 22,500	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 176,644
Total Direct Costs													
Customer Contribution		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Indirect Costs													
Indirect Costs													
Non-Productive(BNP)													
Product Development(PDP)													
General Services(GS)													
MISC(DMS)													
EM/OPS (BIL, BIL, BIL)													
AS&E(DMS)													
Lobby Staff													
Station Test and BIL(BIL)													
Vehicles(DMS)													
Total Indirect Costs		\$ -	\$ 17,664	\$ 2,250	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 20,164
All Costs (incl. BNP, PDP, GS, MISC, EM/OPS, AS&E, Lobby Staff, Station Test and BIL, Vehicles)		\$ -	\$ 165,508	\$ 24,750	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 196,812
TOTALS		\$ -	\$ 165,508	\$ 24,750	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 196,812

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24			BESS - Remaining TJP								06-Mar-24 15:55						
Activity ID	Activity Name	Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources		Predecessors	Successors	2024	2025	2026	2027	2028	
Project Lead: Adam Kelley Adam S. Kelley			1037d	15-Jun-21 A	09-Mar-28												
80089926 Hyde Park BESS - Battery Energy Storage System Solution 03-06-24			1037d	15-Jun-21 A	09-Mar-28												
80089926.01 PROJECT PLANNING & FUNDING			686d	15-Jun-21 A	05-Nov-26												
PPF1000	Project Planning & Funding	686d	15-Jun-21 A	05-Nov-26	In Progress												
80089926.01.12 T-Minus Milestones			661d	29-Mar-24	05-Nov-26												
AT2605	Preliminary Design / Engineering Completed	0d		04-Jun-25	Not Started	ENG-PR	ENG.ST.SS			DE2120, DE2100, DE2140, DE2060, DE2040, DE2160		◆ Preliminary Design / Engineering Completed					
AT1811	Detail Design/Engineering Completed	0d		18-Dec-25	Not Started	ENG-DT	ENG.ST.SS			COM1000, PC1110, UG1280, OH1430, EE1240, EE1220		◆ Detail Design/Engineering Completed					
AT0805	Final Engineering Issued for Construction W/BOM	0d		09-Jan-26	Not Started	ENG-IFC	ENG.ST.SS			COM1030, PC1330, UG1370, OH1690, EE1240, EE1250	ISD	◆ Final Engineering Issued for Construction					
80089926.01.12.01 Civil T-Minus Milestones			930d	29-Mar-24	15-Oct-26												
AT3001	Scope Document Approved	0d		29-Mar-24	Not Started	ENG-SC	ENG.ST.SS			PE1720, ATCGE1010	AT00	◆ Scope Document Approved					
BT2615	Civil Preliminary Design Complete	0d		08-Apr-25	Not Started	ENG-PR	ENG.ST.SS			DE2160	BD1010, PP1000, PHZ1010	◆ Civil Preliminary Design Complete					
AT1030	ISO I.3.9 Approval	0d		30-May-25	Not Started		ENG.SYS.I			ISO1190	AT00	◆ ISO I.3.9 Approval					
BT2625	Civil Detail Design Complete	0d		18-Dec-25	Not Started	ENG-DT	ENG.ST.SS			EE1330	AT00	◆ Civil Detail Design Complete					
AT0805C	IFC - Civil (BG)	0d		09-Jan-26	Not Started	AG-IFC	ENG.ST.SS			EE1350	AT00	◆ IFC - Civil (BG)					
AT0401	Civil Long Lead Items Delivered	0d		19-Feb-26	Not Started		PURC			CMAT1010	AT00	◆ Civil Long Lead Items Delivered					
AT0814	Civil Construction Contracts Awarded - Civil BG	0d		17-Mar-26	Not Started		PURC			BCN2150	AT00	◆ Civil Construction Contracts Awarded					
AT0203	Civil Constr Auth Notice (CAN) completed [go/ no go]	0d		25-Jun-26	Not Started		PM				AT00, BCN2090	◆ Civil Constr Auth Notice (CAN) completed					
AT00	T-00 Construction Start Civil	0d	26-Jun-26*		Not Started		PM			AT0401, AT0203, AT0805C, BT2625, AT0814, AT0801, AT1030	BCN2090	◆ T-00 Construction Start Civil					
AT0801	Environmental & Local Permits Received	0d		15-Oct-26	Not Started		PM			PP1000, BD1000, BD1010	AT00	◆ Environmental & Local Permits Received					
80089926.01.12.02 Electrical T-Minus Milestones			533d	20-May-25	05-Nov-26												
BT2605	Electrical Preliminary Design Complete	0d		20-May-25	Not Started	ENG-PR	ENG.ST.SS			DE2040	BT00, BD1010, PP1000	◆ Electrical Preliminary Design Complete					
BT2404	Assign Sta Tech (Comissioning) Engineer	0d		20-May-25	Not Started		ENG.ST.SL			DE2040	CT00	◆ Assign Sta Tech (Comissioning) Engineer					
BT1811	Electrical Detail Design Complete	0d		17-Oct-25	Not Started	ENG-DT	ENG.ST.SS			EE1210	BT00	◆ Electrical Detail Design Complete					
BT0805E	IFC - Electrical (AG)	0d		07-Nov-25	Not Started	AG-IFC	ENG.ST.SS			EE1310	BT00	◆ IFC - Electrical (AG)					
BT0814	Electrical Construction Contracts Awarded - Electrical AG	0d		26-Dec-25	Not Started		PURC			BCN2050	BT00	◆ Electrical Construction Contracts Awarded					
BT0801	Environmental & Local Permits Received	0d		15-Oct-26	Not Started		PM			PRE1010, PP1000, BD1000, BD1010	BT00	◆ Environmental & Local Permits Received					
BT0203	Electrical Constr Auth Notice (CAN) completed [go/ no go]	0d		04-Nov-26	Not Started		PM				BT00	◆ Electrical Constr Auth Notice (CAN) completed					
BT0401	Electrical Long Lead Items Delivered	0d		04-Nov-26	Not Started		PURC			EMAT1230	BT00	◆ Electrical Long Lead Items Delivered					
BT00	T-00 Construction Start Electrical	0d	05-Nov-26		Not Started		PM			BT0401, BT0203, BT0805E, BT1811, PHZ1010, BT2605	BCN1050, BCN1030, ECN1040	◆ T-00 Construction Start Electrical					
80089926.01.12.06 P&C T-Minus Milestones			390d	22-Apr-25	29-Oct-26												
Actual Work Critical Remaining Work Summary Remaining Work Milestone																	
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Activity ID		Activity Name				Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources		Predecessors	Successors	2024		2025		2026		2027		2028	
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Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024	2025	2026	2027	2028				
			80089926.04.03.02 Utility Commission		200d	01-Sep-23 A	09-Dec-24						09-Dec-24, 80089926.04.03.02 : Utility Commission								
			UC1150	GMAC Review Period	3d	01-Sep-23 A	28-Feb-24	In Progress		PM	Adam S. Kelley	UC1070	UC1080	GMAC Review Period							
			UC1080	File ESMP with DPU	47d	29-Feb-24	03-May-24	Not Started		PM	Adam S. Kelley	UC1150	UC1120	File ESMP with DPU							
			UC1120	DPU Review and Approval (advisory only - no successor required)	150d	06-May-24	09-Dec-24	Not Started		PM	Adam S. Kelley	UC1080		DPU Review and Approval (advisory only - no suc							
			80089926.04.3 Local Approvals And Permits		400d	26-Mar-25	15-Oct-26							15-Oct-26, 80089926.04.3 : Local Approvals And Permits							
			80089926.04.3.01 Building Department		125d	26-Mar-25	19-Sep-25							19-Sep-25, 80089926.04.3.01 : Building Department							
			BD1000	Permit(s) for Bldg; St Opening; Hwy xing; RR xing; Other	125d	26-Mar-25	19-Sep-25	Not Started		PM	Adam S. Kelley	PP1000, DE2150	PHZ1000, BT0801, FINCO1000, AT0801	Permit(s) for Bldg; St Opening; Hwy xi							
			80089926.04.3.09 Special Use Article 80		320d	18-Jul-25	15-Oct-26							15-Oct-26, 80089926.04.3.09 : Special Use Article 80							
			BD1010	Local Siting Review and Approval	320d	18-Jul-25*	15-Oct-26	Not Started		PM	Adam S. Kelley	BT2605, BT2615, DT2605, ET2605	BT0801, FINCO1000, AT0801, BCN2090	Local Siting Review a							
			80089926.05 ENGINEERING DETAILED / IFC DESIGN		151d	31-Jul-22 A	09-Jan-26							09-Jan-26, 80089926.05 : ENGINEERING DETAILED / IFC DESIGN							
			ENG1000	Engineering Detailed / IFC Design - WBS Summary	151d	31-Jul-22 A	09-Jan-26	In Progress						Engineering Detailed / IFC Desig							
			80089926.05.01 Civil Engineering		92d	28-Aug-25	09-Jan-26							09-Jan-26, 80089926.05.01 : Civil Engineering							
			80089926.05.01.01 Civil Engineering Detailed Design		35d	29-Oct-25	18-Dec-25							18-Dec-25, 80089926.05.01.01 : Civil Engineering Detailed Design							
			EE1320	Engineer Develop Civil Substation Detail Design	25d	29-Oct-25	04-Dec-25	Not Started		ENG.ST.SS	Shodhan Patel	DE2160, CV1030, CV1040, PF1010, FF1000	EE1330	Engineer Develop Civil Substation							
			EE1330	ES Engineering Review Civil Substation Detail Design	10d	05-Dec-25	18-Dec-25	Not Started	ES-DTL	ENG.ST.SS	Shodhan Patel	EE1320	EE1340, BT2625, AT1811	ES Engineering Review Civil Subs							
			80089926.05.01.1 Civil Engineering Pre-IFC Design		42d	28-Aug-25	28-Oct-25							28-Oct-25, 80089926.05.01.1 : Civil Engineering Pre-IFC Design							
			CV1010	Notifications for Soil Borings and Test Pits	10d	28-Aug-25	11-Sep-25	Not Started	V-DTL	ENG.ST.SS	Shodhan Patel	PI1050, PF1010, PF1010, FF1000	CV1020	Notifications for Soil Borings and Test							
			CV1020	Perform Site - Soil Borings and Test Pits	10d	12-Sep-25	25-Sep-25	Not Started	V-DTL	ENG.ST.SS	Shodhan Patel	CV1010	CV1030, CV1040	Perform Site - Soil Borings and Test P							
			CV1030	Receive Geotech Report	22d	26-Sep-25	28-Oct-25	Not Started	V-DTL	ENG.ST.SS	Shodhan Patel	CV1020	EE1320	Receive Geotech Report							
			CV1040	Receive Soil Report	22d	26-Sep-25	28-Oct-25	Not Started	V-DTL	ENG.ST.SS	Shodhan Patel	CV1020	EE1320	Receive Soil Report							
			80089926.05.01.02 Civil Engineering IFC Design		15d	19-Dec-25	09-Jan-26							09-Jan-26, 80089926.05.01.02 : Civil Engineering IFC Design							
			EE1340	Engineer Develop Civil Substation IFC	10d	19-Dec-25	02-Jan-26	Not Started		ENG.ST.SS	Shodhan Patel	EE1330	EE1350	Engineer Develop Civil Substation							
			EE1350	ES Engineering Review & Issue Civil Substation IFC	5d	05-Jan-26	09-Jan-26	Not Started	ES-IFC	ENG.ST.SS	Shodhan Patel	EE1340	AT0805C, BCN2100, AT0805	ES Engineering Review & Issue C							
			80089926.05.02 Electrical Engineering		50d	28-Aug-25	07-Nov-25							07-Nov-25, 80089926.05.02 : Electr							
			CSD1000	PE Initiate CASCADE Forms (Appendix A) for Construction Package	5d	03-Nov-25	07-Nov-25	Not Started	CASCADE	ENG.ST.SS	Shodhan Patel	EE1210, EE1310	CSD1050	PE Initiate CASCADE Forms (Appe							
			80089926.05.02.01 Electrical Engineering Detailed Design		35d	28-Aug-25	17-Oct-25							17-Oct-25, 80089926.05.02.01 : Elec							
			EE1200	Engineer Develop Electrical Substation Detail Design	25d	28-Aug-25	02-Oct-25	Not Started		ENG.ST.SS	Shodhan Patel	DE1900, DE2040, FF1000	EE1210, OPERM1000	Engineer Develop Electrical Substatio							
			EE1210	ES Engineering Review Electrical Substation Detail Design	10d	03-Oct-25	17-Oct-25	Not Started	ES-DTL	ENG.ST.SS	Shodhan Patel	EE1200	A3360, AT1811, BCN2000, BT1811, CSD1000, EE1200	ES Engineering Review Electrical Su							
			80089926.05.02.02 Electrical Engineering Pre-IFC Design		15d	20-Oct-25	07-Nov-25							07-Nov-25, 80089926.05.02.02 : Ele							
			EE1300	Engineer Develop Electrical Substation IFC	10d	20-Oct-25	31-Oct-25	Not Started		ENG.ST.SS	Shodhan Patel	EE1210	EE1310	Engineer Develop Electrical Substat							
			EE1310	ES Engineering Review & Issue Electrical Substation IFC	5d	03-Nov-25	07-Nov-25	Not Started	ES-IFC	ENG.ST.SS	Shodhan Patel	EE1300	AT0805, BT0805E, CSD1000, APP1000	ES Engineering Review & Issue Ele							
			80089926.05.06 P&C Engineering		70d	28-Aug-25	09-Dec-25							09-Dec-25, 80089926.05.06 : P&C Engineering							
			OP1010	P&C Validation of PAF's Initial Assessment Go/No-Go	2d	28-Aug-25	29-Aug-25	Not Started		ENG.ST.P&	P&C Engineer	OP241000, PF1010, FF1000	OP1020	P&C Validation of PAF's Initial Assessn							
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Activity ID		Activity Name	Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024	2025	2026	2027	2028				
	OP1020	P&C Updates OP-24 Appendix B	4d	02-Sep-25	05-Sep-25	Not Started		ENG.ST.P&C	P&C Engineer	OP1010	OP1030						P&C Updates OP-24 Appendix B			
	OP1030	P&C Submits OP-24 Appendix B to ISO-NE	1d	08-Sep-25	08-Sep-25	Not Started	OP-24	ENG.ST.P&C	P&C Engineer	OP1020	ET0901						P&C Submits OP-24 Appendix B to ISO-NE			
	CSD1030	PE Initiate CASCADE Forms (Appendix A) for Construction Package	5d	03-Nov-25	07-Nov-25	Not Started	CASCADE	ENG.ST.P&C	P&C Engineer	PC1330, PC1110	CSD1050						PE Initiate CASCADE Forms (Appendix A) for Construction Package			
	80089926.05.06.01 P&C Engineering Detailed Design		35d	28-Aug-25	17-Oct-25													17-Oct-25, 80089926.05.06.01: P&C Engineering Detailed Design		
	PC1100	Engineer Develop P&C Detail Design	25d	28-Aug-25	02-Oct-25	Not Started		ENG.ST.P&C	P&C Engineer	DE2100, FF1000, FF1000, PF1010	PC1110, OPERM1000						Engineer Develop P&C Detail Design			
	PC1110	ES Engineering Review P&C Detail Design	10d	03-Oct-25	17-Oct-25	Not Started	ES-DTL	ENG.ST.P&C	P&C Engineer	PC1100	AT1811, ECN2000, CSD1030, PC1320, FT1011						ES Engineering Review P&C Detail Design			
	80089926.05.06.02 P&C Engineering Pre-IFC Design		35d	20-Oct-25	09-Dec-25													09-Dec-25, 80089926.05.06.02: P&C Engineering Pre-IFC Design		
	80089926.05.06.02.1 P&C Relay Settings Development		20d	10-Nov-25	09-Dec-25													09-Dec-25, 80089926.05.06.02.1: P&C Relay Settings Development		
	PC1400	Complete P&C Relay Settings	20d	10-Nov-25	09-Dec-25	Not Started	ES-RSE	ENG.ST.P&C	P&C Engineer	PC1330	ET0404						Complete P&C Relay Settings			
	80089926.05.06.02.03 P&C Engineering IFC Design		15d	20-Oct-25	07-Nov-25													07-Nov-25, 80089926.05.06.02.03: P&C Engineering IFC Design		
	PC1320	Engineer Develop P&C IFC	10d	20-Oct-25	31-Oct-25	Not Started		ENG.ST.P&C	P&C Engineer	PC1110	PC1330						Engineer Develop P&C IFC			
	PC1330	ES Engineering Review & Issue P&C IFC	5d	03-Nov-25	07-Nov-25	Not Started	ES-IFC	ENG.ST.P&C	P&C Engineer	PC1320	AT0805, A-DFR1010, A-DFR1060, ET0805, PC1400, CSD1030						ES Engineering Review & Issue P&C IFC			
	80089926.05.07 Telecom Engineering		114d	05-Jun-25	17-Nov-25													17-Nov-25, 80089926.05.07: Telecom Engineering		
	CSD1040	PE Initiate CASCADE Forms (Appendix A) for Construction Package	5d	10-Nov-25	17-Nov-25	Not Started	CASCADE	ENG.ST.SL	Bechir El Khoury Hanna	COM1000, COM1030	CSD1050						PE Initiate CASCADE Forms (Appendix A) for Construction Package			
	80089926.05.07.01 Telecomm Engineering Detailed Design		35d	28-Aug-25	17-Oct-25													17-Oct-25, 80089926.05.07.01: Telecomm Engineering Detailed Design		
	COM1040	Engineer Develop Communications Detail Design	25d	28-Aug-25	02-Oct-25	Not Started		ENG.ST.SL	Demetrios A.Sakellaris	DE2120, FF1000, PF1010	COM1000						Engineer Develop Communications Detail Design			
	COM1000	ES Engineering Review Communications Detail Design	10d	03-Oct-25	17-Oct-25	Not Started	ES-DTL	ENG.ST.SL	Bechir El Khoury Hanna	COM1040	COM1050, AT1811, FCN2000, CSD1040, COMMAT1000						ES Engineering Review Communications Detail Design			
	80089926.05.07.02 Telecomm Engineering Pre-IFC Design		109d	05-Jun-25	07-Nov-25													07-Nov-25, 80089926.05.07.02: Telecomm Engineering Pre-IFC Design		
	COM1010	Confirm Duct Bank/ Poles/ Inner Duct for Fiber is Complete	21d	05-Jun-25	03-Jul-25	Not Started		ENG.ST.SL	Bechir El Khoury Hanna	DE2120	COM1020						Confirm Duct Bank/ Poles/ Inner Duct for Fiber is Complete			
	COM1020	Comm Constructability Review	21d	07-Jul-25	04-Aug-25	Not Started		ENG.ENGF	Field Communication North	COM1010	FT00, FT0805M, FT0203, FCN1010, FCN1020						Comm Constructability Review			
	COM1050	Engineer Develop Communications IFC	10d	20-Oct-25	31-Oct-25	Not Started		ENG.ST.SL	Bechir El Khoury Hanna	COM1000	COM1030						Engineer Develop Communications IFC			
	COM1030	ES Engineering Review & Issue Communications IFC	5d	03-Nov-25	07-Nov-25	Not Started	ES-IFC	ENG.ST.SL	Bechir El Khoury Hanna	COM1050	AT0805, CSD1040						ES Engineering Review & Issue Communications IFC			
80089926.06 PROCUREMENT LONG LEAD MATERIAL		305d	28-Aug-25	04-Nov-26													04-Nov-26, 80089926.06: PROCUREMENT LONG LEAD MATERIAL			
PLLM1000	Procurement Long Lead Material	305d	28-Aug-25	04-Nov-26	Not Started												Procurement Long Lead Material			
80089926.06.01 Civil Long Lead Material		121d	28-Aug-25	19-Feb-26													19-Feb-26, 80089926.06.01: Civil Long Lead Material			
CMAT1020	Civil (BG) Material DR Complete	0d		28-Aug-25	Not Started		ENG.ST.SS		CMAT1000, PF1010, EOC1040	CMAT1010						Civil (BG) Material DR Complete				
CMAT1000	Prepare Civil Material DR	1d	28-Aug-25	28-Aug-25	Not Started	DR-M	ENG.ST.SS	Shodhan Patel	FF1000	CMAT1020, CMAT1010						Prepare Civil Material DR				
CMAT1010	Civil Material Fab & Delivery	120d	29-Aug-25	19-Feb-26	Not Started	M-DELIV	PURC	Purchasing Department	CMAT1000, CMAT1020, PF1010	AT0401, BCN2090						Civil Material Fab & Delivery				
80089926.06.02 Material Procurement - Batteries & Inverters		305d	28-Aug-25	04-Nov-26													04-Nov-26, 80089926.06.02: Material Procurement - Batteries & Inverters			
EMAT1100	Issue Spec. - Material Procurement - Batteries & Inverters	15d	28-Aug-25	18-Sep-25	Not Started		ENG.ST.SS	Shodhan Patel	DE2040, FF1000, EOC1040	EMAT1110, EMAT1210, EMAT1180						Issue Spec. - Material Procurement - Batteries & Inverters				
EMAT1180	Material Vendor Submits Shop Drawings - Material Procurement - Batteries & Inverters	10d	19-Sep-25	02-Oct-25	Not Started		PURC	Purchasing Department	EMAT1100	EMAT1190						Material Vendor Submits Shop Drawings - Material Procurement - Batteries & Inverters				
EMAT1190	ES Reviews Shop Drawings - Material Procurement - Batteries & Inverters	10d	03-Oct-25	17-Oct-25	Not Started		ENG.ST.SS	Shodhan Patel	EMAT1180	EMAT1200						ES Reviews Shop Drawings - Material Procurement - Batteries & Inverters				
Actual Work		Critical Remaining Work		Summary																
Remaining Work		Milestone																		
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Activity ID		Activity Name	Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024	2025	2026	2027	2028				
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Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024		2025		2026		2027		2028

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Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024		2025		2026		2027		2028		
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Activity ID		Activity Name	Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024	2025	2026	2027	2028			
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Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024	2025	2026	2027	2028			

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24			BESS - Remaining TJP								06-Mar-24 15:56												
Activity ID	Activity Name	Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources		Predecessors	Successors	2024	2025	2026	2027	2028							
	EMAT1130	N/R 12-7-23 MR RFP - Long Lead Item #1	0d	17-Dec-25 A	31-Dec-25 A	Comple		PURC	Purchasing Department	EMAT1120	EMAT1135					N/R 12-7-23 MR RFP - Long Lea							
	COMMAT1040	N/R 12-7-23 MR RFP - GE JMUX	0d	17-Dec-25 A	31-Dec-25 A	Comple		PURC	Purchasing Department	COMMAT1030	COMMAT1050					N/R 12-7-23 MR RFP - GE JMUX							
	PCMAT1050	N/R 12-7-23 MR RFP - Long Lead Item #1	0d	17-Dec-25 A	31-Dec-25 A	Comple		PURC	Purchasing Department	PCMAT1040	PCMAT1060					N/R 12-7-23 MR RFP - Long Lea							
	DT0814	N/R 12-7-23 MR Construction Contracts Awarded -	0d		31-Dec-25 A	Comple		PURC			DT00					N/R 12-7-23 MR Construction Co							
	DMAT1030	N/R 12-7-23 MR RFP - Long Lead Item #1	0d	17-Dec-25 A	31-Dec-25 A	Comple		PURC	Purchasing Department	DMAT1020	DMAT1040					N/R 12-7-23 MR RFP - Long Lea							
	COMMAT1050	N/R 12-7-23 MR ES Selects Material Vendor - GE JMUX	0d	01-Jan-26 A	14-Jan-26 A	Comple		PURC	Purchasing Department	COMMAT1040	COMMAT1060					N/R 12-7-23 MR ES Selects Mat							
	PCMAT1060	N/R 12-7-23 MR ES Provides Eng Specs.To Bidders - Long Lead Item #1	0d	01-Jan-26 A	14-Jan-26 A	Comple		PURC	Purchasing Department	PCMAT1050	PCMAT1070					N/R 12-7-23 MR ES Provides Er							
	DMAT1040	N/R 12-7-23 MR ES Provides Eng Specs.To Bidders - Long Lead Item #1	0d	01-Jan-26 A	14-Jan-26 A	Comple		PURC	Purchasing Department	DMAT1030	DMAT1050					N/R 12-7-23 MR ES Provides Er							
	A-DFR1130	Procure DFR Mat N/R	20d	22-Dec-25	19-Jan-26	Not Started		PURC	Purchasing Department	A-DFR1120						Procure DFR Mat N/R							
	DT0203	N/R 12-7-23 MR Constr Auth Notice (CAN) completed [go/ no go]	0d		20-Jan-26 A	Comple		PM			DT00					N/R 12-7-23 MR Constr Auth No							
	DT00	N/R 12-7-23 MR T-00 Construction Start Distribution (Outage Start if needed)	0d	21-Jan-26 A		Comple		PM		DT1811, DT2605, DT0805UG, DT0814, DT0602, DT0404, DT0303	DCN1030, DCN1020, DCN1010					N/R 12-7-23 MR T-00 Constructi							
	EMAT1135	N/R 12-7-23 MR ES Provides Eng Specs.To Bidders - Long Lead Item #1	0d	01-Jan-26 A	21-Jan-26 A	Comple		PURC	Purchasing Department	EMAT1130	EMAT1140					N/R 12-7-23 MR ES Provides Er							
	COMMAT1060	N/R 12-7-23 MR ES Issues P.O. - GE JMUX	0d	15-Jan-26 A	28-Jan-26 A	Comple		PURC	Purchasing Department	COMMAT1050						N/R 12-7-23 MR ES Issues P.O.							
	PCMAT1070	N/R 12-7-23 MR Bid Period - Long Lead Item #1	0d	15-Jan-26 A	28-Jan-26 A	Comple		PURC	Purchasing Department	PCMAT1060	PCMAT1080					N/R 12-7-23 MR Bid Period - Lo							
	DMAT1050	N/R 12-7-23 MR Bid Period - Long Lead Item #1	0d	15-Jan-26 A	28-Jan-26 A	Comple			Purchasing Department	DMAT1040	DMAT1060					N/R 12-7-23 MR Bid Period - Lo							
	PCMAT1080	N/R 12-7-23 MR ES Reviews Submitted Bids - Long Lead Item #1	0d	29-Jan-26 A	11-Feb-26 A	Comple		PURC	Purchasing Department	PCMAT1070	PCMAT1090					N/R 12-7-23 MR ES Reviews S							
	DMAT1060	N/R 12-7-23 MR ES Reviews Submitted Bids - Long Lead Item #1	0d	29-Jan-26 A	11-Feb-26 A	Comple		PURC	Purchasing Department	DMAT1050	BT0408, DMAT1070					N/R 12-7-23 MR ES Reviews S							
	EMAT1140	N/R 12-7-23 MR Bid Period - Long Lead Item #1	0d	22-Jan-26 A	18-Feb-26 A	Comple		PURC	Purchasing Department	EMAT1135	EMAT1150					N/R 12-7-23 MR Bid Period - L							
	PCMAT1090	N/R 12-7-23 MR ES Selects Material Vendor - Long Lead Item #1	0d	12-Feb-26 A	25-Feb-26 A	Comple		PURC	Purchasing Department	PCMAT1080	PCMAT1100					N/R 12-7-23 MR ES Selects M							
	DMAT1070	N/R 12-7-23 MR ES Selects Material Vendor - Long Lead Item #1	0d	12-Feb-26 A	25-Feb-26 A	Comple		PURC	Purchasing Department	DMAT1060	DMAT1080					N/R 12-7-23 MR ES Selects M							
	EMAT1150	N/R 12-7-23 MR ES Reviews Submitted Bids - Long Lead Item #1	0d	19-Feb-26 A	04-Mar-26 A	Comple		PURC	Purchasing Department	EMAT1140	BT0408, EMAT1160					N/R 12-7-23 MR ES Reviews S							
	DT0602	N/R 12-7-23 MR Units of Property issued to Plant Accounting	0d		06-Mar-26 A	Comple	PM	PM		FINCO1010	DT00					N/R 12-7-23 MR Units of Prop							
	PCMAT1100	N/R 12-7-23 MR ES Issues P.O. / ES Approves CR - Long Lead Item #1	0d	26-Feb-26 A	11-Mar-26 A	Comple		PURC	Purchasing Department	PCMAT1090						N/R 12-7-23 MR ES Issues P.							
	DMAT1080	N/R 12-7-23 MR ES Issues P.O. / ES Approves CR - Long Lead Item #1	0d	26-Feb-26 A	11-Mar-26 A	Comple		PURC	Purchasing Department	DMAT1070	DMAT1090					N/R 12-7-23 MR ES Issues P.							
	EMAT1160	N/R 12-7-23 MR ES Selects Material Vendor - Long Lead Item #1	0d	05-Mar-26 A	18-Mar-26 A	Comple		PURC	Purchasing Department	EMAT1150	EMAT1170					N/R 12-7-23 MR ES Selects M							
	DMAT1090	N/R 12-7-23 MR Material Vendor Submits Shop Drawings - Long Lead Item #1	0d	12-Mar-26 A	25-Mar-26 A	Comple		PURC	Purchasing Department	DMAT1080	DMAT1100					N/R 12-7-23 MR Material Ven							
	EMAT1170	N/R 12-7-23 MR ES Issues P.O. / ES Approves CR - Long Lead Item #1	0d	19-Mar-26 A	01-Apr-26 A	Comple		PURC	Purchasing Department	EMAT1160						N/R 12-7-23 MR ES Issues P							
	DMAT1100	N/R 12-7-23 MR ES Reviews Shop Drawings - Long Lead Item #1	0d	26-Mar-26 A	08-Apr-26 A	Comple		ENG.ST.LII	Michael Hickey	DMAT1090	DMAT1110					N/R 12-7-23 MR ES Reviews							
Actual Work		Critical Remaining Work	Summary		Page 15 of 16						TASK filter: EMA REPORT - Remaining TJP.												
Remaining Work		Milestone							© Oracle Corporation														

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24			BESS - Remaining TJP								06-Mar-24 15:56											
Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024		2025		2026		2027		2028	
	DMAT1110	N/R 12-7-23 MR ES Approves Shop Drawings - Long Lead Item #1		0d	09-Apr-26 A	22-Apr-26 A	Comple		ENG.ST.LII	Michael Hickey	DMAT1100	DMAT1120	N/R 12-7-23 MR ES Approve									
	DMAT1120	N/R 12-7-23 MR Material Vendor Manufactures Long Lead Item #1		0d	23-Apr-26 A	06-May-26 A	Comple		PURC	Purchasing Department	DMAT1110	DMAT1130	N/R 12-7-23 MR Material Ve									
	DMAT1130	N/R 12-7-23 MR Material Vendor Delivers Long Lead Item #1		0d	07-May-26 A	20-May-26 A	Comple	M-DELI	PURC	Purchasing Department	DMAT1120	DT0401	N/R 12-7-23 MR Material V									